

Problem 1

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;

    int s =1;

    while(s <= n)
    {
        i++;
        s += i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2
3  int count=0;
4  void function(int n)
5
6  {
7      int i=1;
8      count++;
9      int s=1;
10     count++;
11
12     while(s<=n)
13     {
14         count++;
15         i++;
16         count++;
17         s +=i;
18         count++;
19     }
20     count++;
21     printf("%d",count);
22 }
23 int main()
24 {
25     int n;
26     scanf("%d",&n);
27     function(n);
28 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Problem 2

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("**");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("**");
                printf("**");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int count=0;
3
4  void func(int n)
5  {
6      if(n==1)
7      {
8          count++;
9          printf("");
10     }
11     else
12     {
13         count++;
14         for(int i=1;i<=n;i++)
15         {
16             count++;
17             for(int j=1;j<=n;j++)
18             {
19
20                 count++;
21                 //printf("");
22                 count++;
23                 //printf("");
24                 count++;
25                 break;
26             }
27             count++;
28         }
29         count++;
30     }
31     printf("%d",count);
32 }
33 int main()
34 {
35     int n;
36     scanf("%d",&n);
37     func(n);
38
39 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Problem 3:

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {  
  {  
    for (i = 1; i <= num; ++i)  
    {  
      if (num % i == 0)  
      {  
        printf("%d ", i);  
      }  
    }  
  }  
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int count=0;
3
4  void factor(int num)
5  {
6      for(int i=1;i<=num;++i)
7      {
8          count++;
9          if(num%i==0)
10         {
11             count++;
12         }
13         count++;
14     }
15     count++;
16     printf("%d",count);
17 }
18
19 int main()
20 {
21     int n;
22     scanf("%d",&n);
23     factor(n);
24 }
25
26
27 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Problem 4

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int count=0;
3
4  void function(int n)
5  {
6      int c=0;
7      count++;
8      for(int i=n/2;i<n;i++)
9      {
10         count++;
11         for(int j=1;j<n;j=2*j)
12         {
13             count++;
14             for(int k=1;k<n;k=k*2)
15             {
16                 count++;
17                 c++;
18                 count++;
19             }
20             count++;
21         }
22         count++;
23     }
24     count++;
25     printf("%d",count);
26 }
27 int main()
28 {
29     int n;
30     scanf("%d",&n);
31     function(n);
32 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Problem 5

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;

    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include<stdio.h>
2 int count=0;
3
4 void reverse(int n)
5 {
6     int rev=0,remainder;
7     count++;
8     while(n!=0)
9     {
10         count++;
11         remainder = n%10;
12         count++;
13         rev = rev*10+remainder;
14         count++;
15         n/=10;
16         count++;
17     }
18     count++;
19     count++;
20     printf("%d",count);
21 }
22 int main()
23 {
24     int n;
25     scanf("%d",&n);
26     reverse(n);
27 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓